

Adjuvant Radiation Therapy Improves Patient Survival in Early-Stage Merkel Cell Carcinoma: A 15-Year Single-Institution Study

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Objectives/Hypothesis: Merkel cell carcinoma (MCC) is a rare and aggressive neuroendocrine neoplasm of the skin. Growing evidence supports the benefit of postoperative adjuvant radiation therapy (RT) for locoregional control, but whether it improves overall survival (OS) has been debated. Our objective was to compare the OS of MCC patients who received postoperative RT with those who received surgery alone.

Study Design: Retrospective case series.

Methods: Cases of MCC between 2001 and 2016 at the University of California, Los Angeles Health System were reviewed. We identified 87 unique cases of MCC. Among the patients, 74% were identified as male and 26% as female. The average age at diagnosis was 71.2 years. The median survival was 48.0 months. The OS of all the patients at 2 years, 5 years, and 10 years was 54%, 46%, and 26%, respectively. Univariate analysis showed that stage, T stage, N stage, and M stage were significant determinants of OS. The inclusion of RT was not found to be a determinant; however, when restricting the analysis to early-stage MCC (stages I and II), postoperative adjuvant RT was associated with significantly improved OS. A Cox regression model confirmed that inclusion of RT was an independent prognosticator of OS even when controlled for overall stage and negative margin status. The small sample size and retrospective nature of this study limit its statistical power.

Conclusions: MCC is an aggressive tumor with a poor prognosis for survival especially in elderly patients. In this study, we found that RT during early-stage MCC improves OS. Prospective randomized control trials are necessary to validate the observed benefit for MCC patients.

Key Words: Merkel cell carcinoma, surgical oncology, head and neck cancer, radiation therapy.

Level of Evidence: 4

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INTRODUCTION

Merkel cell carcinoma (MCC) is a rare and aggressive neuroendocrine tumor with high mortality rate. It was first described by Toker as a “trabecular carcinoma of the skin” in 1972.¹ Since the initial discovery, the incidence of MCC has grown three-fold, with a recent estimate of 0.2 to 0.45 cases per 100,000.^{2,3} The exact cause of MCC is unclear. However, risk factors associated with the development of MCC have been proposed, including chronic sun exposure, immunosuppression, a history of

malignancy, and exposure to Merkel cell polyomavirus.^{4–6} More specifically, the risk for human immunodeficiency virus (HIV) patients and heterotopic organ recipients is 10 times greater.^{7,8}

The mainstay treatment for MCC is surgical excision of the lesion followed by adjuvant radiation therapy (RT), chemotherapy, or a combination of both (CRT). The radiosensitive nature of MCC compared to other cutaneous cancers like melanoma has been well characterized in cell line studies.⁹ Primary RT can be considered for patients who are poor surgical candidates in order to achieve definitive or palliative treatment. Chemotherapy is an option for advanced-stage MCC, although evidence is conflicting regarding the optimal duration and timing of adjuvant cytotoxic chemotherapy. Recently, immunotherapy regimens have been explored using pembrolizumab and avelumab; programmed death 1 inhibitors have shown promising results, especially in advanced disease.^{10,11}

The primary objective of this study was to determine the benefit of postoperative adjuvant RT to the primary site in the setting of MCC. Despite the accepted treatment of postoperative adjuvant RT, the overall survival (OS) benefit of RT has been debated. Our results support that postoperative adjuvant RT improves OS, especially in those who present with early-stage MCC.

Additional Supporting Information may be found in the online version of this article.

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TABLE I. Characteristics of Patients With Merkel Cell Carcinoma.	
Age at diagnosis, yr	
Mean	71.18 ± 14.45
Minimum	15
Maximum	98
Characteristic, % (n)	
Sex	
Female	26.0% (20)
Male	74.0% (57)
Immunosuppression, yes, (n = 63)	19.5% (15)
History of prior malignancy, yes, (n = 63)	44.2% (34)

MATERIALS AND METHODS

All cases of MCC observed at Ronald Reagan UCLA Medical Center (UCLA-RR) and Santa Monica UCLA Medical Center (UCLA-SM) to date were retrieved from the medical charts with institutional review board approval. The following primary data were extracted: age, sex, site of primary, negative margin status, presence of nodal or distant metastasis at the time of diagnosis, types of treatments received, and histology. T stage was retrospectively determined when possible using the extent of disease from pathology reports and using tumor size and location according to the classification protocol developed by the American Joint Committee on Cancer, 7th edition. Tumors were staged as T1 for sizes less than 2 cm, T2 for sizes between 2 and 5 cm, T3 for sizes greater than 5 cm, and T4 if primary tumor demonstrated invasion into the surrounding bone, muscle, fascia, or cartilage.¹²

The primary outcome for OS was defined as the time in months from diagnosis to death from any cause. Descriptive statistics were calculated for all variables. The OS curves were calculated using the Kaplan-Meier method, and the differences were formally tested using the log-rank test.¹³ Cox proportional hazard regression models concerning OS were used to assess covariates for their predictive performance. A threshold of $P < .05$ was used to determine statistically significant differences between groups. Covariates were chosen for the multivariate

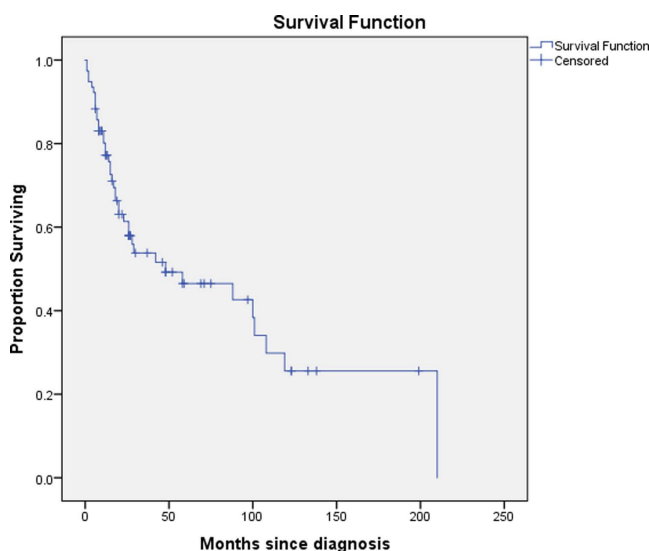


Fig. 1. Kaplan-Meier curve depicting the overall survival of all patients included in this study. [Color figure can be viewed in the online issue, which is available at www.laryngoscope.com.]

TABLE II. Tumor Staging and Anatomical Location of the Primary Lesion.	
Tumor Characteristics	Percentage (No.)
Location	
Head and neck	49.4% (38)
Upper extremities	27.3% (21)
Trunk	6.5% (5)
Lower extremities	16.9% (13)
Margin status, negative	71.1% (27)
Stage	
I or II	5.2% (4)
I	22.1% (17)
II	24.7% (19)
III	41.6% (32)
IV	6.5% (5)
T	
X (unknown primary)	15.6% (12)
1	29.9% (23)
2	37.7% (29)
3	10.4% (8)
4	1.3% (1)
N/A	5.2% (4)
N	
cN0	23.4% (18)
pN0	29.9% (23)
N1a	9.1% (7)
N1b	35.1% (27)
N2	2.6% (2)
M	
0	93.5% (72)
1	6.5% (5)

N/A = not available.

analysis on the basis of factors identified as significant on univariate analysis, as well as the factors that have been known to be related to the decision for radiation treatment (e.g., negative margin status). Efforts were taken to minimize the total number of covariates, thus improving the generalizability of the findings and minimizing the instability in the model.¹⁴ Statistical analyses were performed with SPSS version 23 (IBM Corp., Armonk, NY).

TABLE III. Treatment Rendered to Patients With Merkel Cell Carcinoma.	
Treatment Modality	Percentage (No.)
Surgery alone	23.4% (14)
Surgery + RT	26.0% (20)
Surgery + RT + chemotherapy	27.3% (21)
Surgery + chemotherapy	3.9% (3)
Chemotherapy + RT	1.3% (1)
RT alone	2.6% (2)
Unknown	14.3% (11)

RT = radiation therapy.

TABLE IV.

Univariate Analysis of Predictors for Overall Survival, Calculated From Kaplan-Meier Analysis With Comparisons Performed With the Log-Rank Test.

Characteristic	Overall Survival (Log-Rank <i>P</i>)
Age (by each decade)	.060
Sex	.141
Primary site	.388
Surgery performed	.221
Radiation therapy included	.107 (all cases) .019 (stages I and II)*
Negative margin status	.508
Immunosuppression	.327
History of prior malignancy	.088
Stage	.012*
T	.008*
N	.024*
M	.009*

*Statistical significance ($P < .05$).

RESULTS

Eighty-eight patients with biopsy-proven MCC were seen at UCLA-RR/SM from 2001 to 2016. Data from one chart were excluded from analysis as the chart was a duplicate chart of a preexisting patient. Males accounted for 74% of all the cases (Table I). The mean age at diagnosis was 71.2 years (range, 15–98 years). In terms of their past medical history, 19.5% of the patients had therapeutic immunosuppression or HIV. Slightly less than half (44.2%) had a prior or concurrent malignancy. The OS at 2 years, 5 years, and 10 years was 54%, 46%, and 26%, respectively. Patients in this study had a mean survival of 48 months (Fig. 1), with a 2-year survival and 5-year survival rate of 54% and 46%, respectively.

Regarding the anatomical location, head and neck MCC accounted for 49.4% of the cases. Moreover, 27.3%

cases involved the upper extremities, 6.5% the trunk and 16.9% the lower extremities (Table II). Twenty-three patients were at T1 tumor stage, and 29 patients were at T2 tumor stage; T3 was found in eight patients and T4 in one patient. T staging information was not available for four patients who were excluded in subsequent analyses. The median size of the primary tumor was 2.5 cm. Twelve patients suffered from occult primary malignancy (TX). Regarding locoregional metastases, 29.9% ($n = 23$) of the patients had no pathologic lymph node metastasis (pN0), 44.2% ($n = 34$) had involvement of the lymph nodes (N1a and N1b), and 2.6% ($n = 2$) had tumors that had spread to a nearby lymph node area (N2). Of the patients, 6.5% ($n = 5$) of the patients had distant metastasis (M1).

Next, we categorized the cases by treatments. Among all cases, 23.4% of the patients received surgery alone, 26% received surgery with postoperative adjuvant RT, 27.3% received surgery with postoperative adjuvant CRT, 3.9% received CRT alone, and 2.6% received RT alone (Table III). Complete treatment information was not available for 14.3% of the patients and was thus excluded from further analysis.

Univariate analysis revealed that overall stage, T stage, N stage, and M stage were significant determinants of survival (Table IV). The mean OS is best with stage I disease of 140 months compared to patients with distant metastasis (stage IV) of 8.8 months (Fig. 2A). Age (grouped by each decade), sex, primary site, negative margin status, immunosuppression, and history of prior malignancy were not significant determinants of OS. The inclusion of RT was not found to be a determinant of survival; however, when restricted to early-stage MCC (stages I and II), the addition of RT significantly improved survival ($P = .019$) (Fig. 2B). A Cox regression model revealed that the effect of inclusion of RT (hazard ratio [HR]: 0.237; 95% confidence interval [CI]: 0.066–0.848) was independent of negative surgical margin (HR:

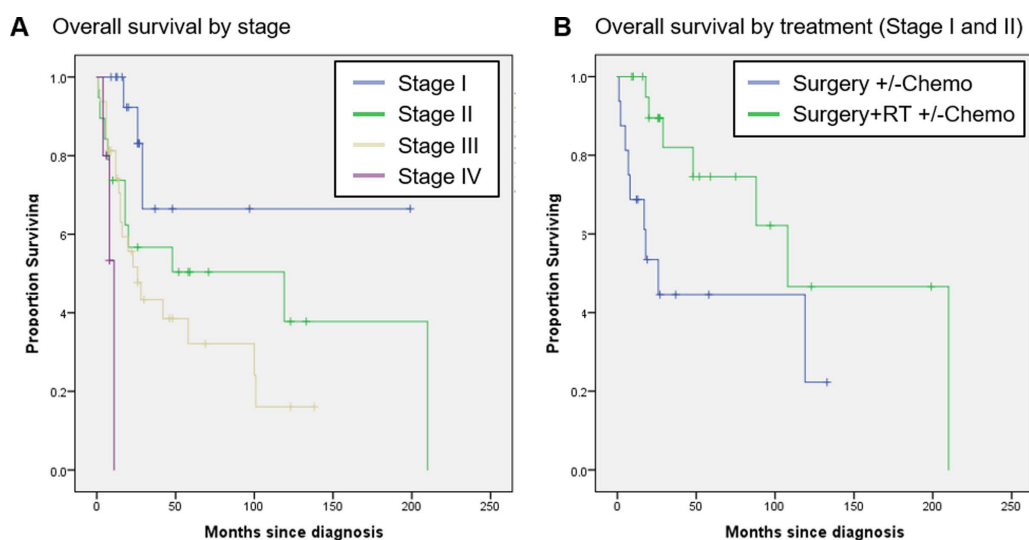


Fig. 2. Kaplan-Meier curve depicting the overall survival of patients with Merkel cell carcinoma categorized by stage (A) and treatment (B, only stages I and II). RT = radiation therapy. [Color figure can be viewed in the online issue, which is available at www.laryngoscope.com.]

TABLE V.
Cox Proportional Hazard Ratio for Overall Survival for Merkel Cell Carcinoma Stages I-II.

Characteristic	Overall Survival	
	HR (95% CI)	P Value
Stages I-II, n = 40		
Radiation therapy included	0.24 (0.07-0.85)	.027*
Overall stage	1.40 (0.61-3.18)	.43
Negative margin status	0.76 (0.22-2.67)	.67

*Statistical significance ($P < .05$).

CI = confidence interval; HR = hazard ratio.

1.320; 95% CI: 0.374-4.652) or stage of the cancer (HR: 1.394; 95% CI: 0.612-3.176) (Table V). The clinicopathologic features of the group that received radiation versus the group that did not were not significantly different when tested with a χ^2 test (see Supporting Table 1 in the online version of this article).

DISCUSSION

MCC is a rare, aggressive cutaneous malignancy with high mortality rate and a propensity to recur despite treatment. In recent decades, a growing body of literature has captured the epidemiology and outcomes of different management options. The demographics of the patients in our study were consistent with those of previously published reports.² Similar to previous studies, MCC most commonly affects older patients in their 70s with a male predominance.

In this study, the head and neck region was the most common anatomical site of the primary lesion (49.4%), slightly higher than Surveillance, Epidemiology, and End Results (SEER) MCC study (29%).^{6,15} We did not find any differences in survival among patients when compared by different anatomical sites which is consistent with the findings of Eich et al.¹⁶ A large portion of the patients had a history of malignancy as well as immunosuppression, which attests for the increased risk of developing MCC with such comorbidities.

Currently accepted treatment for MCC begins with a wide local excision with 1 to 2 cm margins around the primary cancer followed by adjuvant RT or CRT.¹⁷ It has been reported that margin status influences the decision to use postoperative radiation therapy.¹⁸ However, previous studies have reported that margin-negative excision is in fact not associated with disease-specific survival in non-metastatic MCC.^{19,20} Our negative margin status univariate analysis results were consistent with this finding.

The most salient finding in our study is that the addition of postoperative RT was associated with the improved OS of patients in the early stage (Fig. 2B). The literature thus far is inconsistent regarding the survival benefit of adjuvant RT. There has not yet been a prospective randomized trial on postoperative adjuvant RT in the setting of MCC. It has been well accepted that adjuvant RT reduces recurrence.¹⁷ Earlier studies have demonstrated that although postoperative adjuvant RT

can help with locoregional control and disease-free survival, it does not increase OS.^{16,21} In the largest head and neck multicenter retrospective study with 110 patients, Clark and colleagues found that postoperative RT improved survival only in the univariate analysis, but not in the multivariate analysis.²² Interestingly, Clark et al.²² reported OS benefit in stage II MCC. In a separate retrospective analysis by an Australian group, postoperative RT conferred survival benefit in early stage MCC.²⁰ An aggregate-level study that combined all previously reported studies also demonstrated greater OS with postoperative RT.²³ Furthermore, recent studies of the two largest population-based databases, the National Cancer Data Base and SEER, have shown improved OS with adjuvant RT in early-stage MCC.^{24,25}

Although the most salient conclusion in this study stems from our single-institutional experience, which allows for detailed pathologic data and treatment course as well as comorbid conditions, there were some limitations to our study. First, it was not prospective or randomized. Second, the sample size was small, which offers suboptimal statistical power. Age and sex were not found to be independent prognosticators in our analysis, which may be due to the small sample size and inherent bias due to the nature of patients who are referred to a tertiary academic medical center. In addition, about 44% of the patients in surgery with postoperative RT received concurrent chemotherapy. Ideally, these patients would be analyzed in a separate group for a more accurate analysis. Finally, the intent of treatment, whether therapeutic or palliative, was not taken into account, which may have influenced the course of treatment.

CONCLUSION

MCC is an aggressive cutaneous malignancy that most commonly involves the head and neck region. Patients with MCC are usually males in their seventh decade. A large proportion of MCC patients have immunosuppression and prior history of malignancy. Adjuvant postoperative RT was associated with improved OS in early-stage MCC, even when controlled for disease stage and negative margin status. Future prospective trials are recommended to validate these findings and explore the benefit of definitive RT in the management of MCC.

BIBLIOGRAPHY

1. Toker C. Trabecular carcinoma of the skin. *Arch Dermatol* 1972;105:107-110.
2. Agelli M, Clegg LX. Epidemiology of primary Merkel cell carcinoma in the United States. *J Am Acad Dermatol* 2003;49:832-841.
3. Hodgson NC. Merkel cell carcinoma: changing incidence trends. *J Surg Oncol* 2005;89:1-4.
4. Feng H, Shuda M, Chang Y, Moore PS. Clonal integration of a polyomavirus in human Merkel cell carcinoma. *Science* 2008;319:1096-1100.
5. Asgari MM, Sokil MM, Warton EM, Iyer J, Paulson KG, Nghiem P. Effect of host, tumor, diagnostic, and treatment variables on outcomes in a large cohort with Merkel cell carcinoma. *JAMA Dermatol* 2014;150:716-723.
6. Heath M, Jaimes N, Lemos B, et al. Clinical characteristics of Merkel cell carcinoma at diagnosis in 195 patients: the AEIOU features. *J Am Acad Dermatol* 2008;58:375-381.
7. Penn I, First MR. Merkel's cell carcinoma in organ recipients: report of 41 cases. *Transplantation* 1999;68:1717-1721.
8. Engels EA, Frisch M, Goedert JJ, Biggar RJ, Miller RW. Merkel cell carcinoma and HIV infection. *Lancet* 2002;359:497-498.

9. Leonard JH, Ramsay JR, Kearsley JH, Birrell GW. Radiation sensitivity of Merkel cell carcinoma cell lines. *Int J Radiat Oncol Biol Phys* 1995;32:1401–1407.
10. Nghiem PT, Bhatia S, Lipson EJ, et al. PD-1 blockade with pembrolizumab in advanced Merkel-cell carcinoma. *N Engl J Med* 2016;374:2542–2552.
11. Kaufman HL, Russell J, Hamid O, et al. Avelumab in patients with chemotherapy-refractory metastatic Merkel cell carcinoma: a multicentre, single-group, open-label, phase 2 trial. *Lancet Oncol* 2016;17:1374–1385.
12. Edge S, Byrd DR, Compton CC, Fritz AG, Greene FL, Trotti A. AJCC Cancer Staging Handbook: From the AJCC Cancer Staging Manual. New York, NY: Springer; 2011.
13. Kaplan EL, Meier P. Nonparametric estimation from incomplete observations. *J Am Stat Assoc* 1958;53:457–481.
14. Mallen-St Clair J, Arshi A, Tajudeen B, Abemayor E, St John M. Epidemiology and treatment of lacrimal gland tumors: a population-based cohort analysis. *JAMA Otolaryngol Head Neck Surg* 2014;140:1110–1116.
15. Koljonen V. Merkel cell carcinoma. *World J Surg Oncol* 2006;4:7.
16. Eich HT, Eich D, Staar S, et al. Role of postoperative radiotherapy in the management of Merkel cell carcinoma. *Am J Clin Oncol* 2002;25:50–56.
17. Cassler NM, Merrill D, Bichakjian CK, Brownell I. Merkel cell carcinoma therapeutic update. *Curr Treat Options Oncol* 2016;17:36.
18. Rao NG. Review of the role of radiation therapy in the management of Merkel cell carcinoma. *Curr Probl Cancer* 2010;34:108–117.
19. Allen PJ, Bowne WB, Jaques DP, Brennan MF, Busam K, Coit DG. Merkel cell carcinoma: prognosis and treatment of patients from a single institution. *J Clin Oncol* 2005;23:2300–2309.
20. Jabbour J, Cumming R, Scolyer RA, Hruby G, Thompson JF, Lee S. Merkel cell carcinoma: assessing the effect of wide local excision, lymph node dissection, and radiotherapy on recurrence and survival in early-stage disease—results from a review of 82 consecutive cases diagnosed between 1992 and 2004. *Ann Surg Oncol* 2007;14:1943–1952.
21. Veness MJ, Morgan GJ, Gebbski V. Adjuvant locoregional radiotherapy as best practice in patients with Merkel cell carcinoma of the head and neck. *Head Neck* 2005;27:208–216.
22. Clark JR, Veness MJ, Gilbert R, O'Brien CJ, Gullane PJ. Merkel cell carcinoma of the head and neck: is adjuvant radiotherapy necessary? *Head Neck* 2007;29:249–257.
23. Lewis KG, Weinstock MA, Weaver AL, Otley CC. Adjuvant local irradiation for Merkel cell carcinoma. *Arch Dermatol* 2006;142:693–700.
24. Mojica P, Smith D, Ellenhorn JD. Adjuvant radiation therapy is associated with improved survival in Merkel cell carcinoma of the skin. *J Clin Oncol* 2007;25:1043–1047.
25. Bhatia S, Storer BE, Iyer JG, et al. Adjuvant radiation therapy and chemotherapy in Merkel cell carcinoma: survival analyses of 6908 cases from the National Cancer Data Base. *J Natl Cancer Inst* 2016;108(9).